



Machine Learning in Production Planning for Operations

Administrativa

Midsemester grades

M2 due next week Monday

I3 to be released afterward (AI agent/MCP security)

Operations

Fundamentals of Engineering AI-Enabled Systems

Holistic system view: AI and non-AI components, pipelines, stakeholders, environment interactions, feedback loops

Requirements:

System and model goals
User requirements
Environment assumptions
Quality beyond accuracy
Measurement
Risk analysis
Planning for mistakes

Architecture + design:

Modeling tradeoffs
Deployment architecture
Data science pipelines
Telemetry, monitoring
Anticipating evolution
Big data processing
Human-AI design

Quality assurance:

Model testing
Data quality
QA automation
Testing in production
Infrastructure quality
Debugging

Operations:

Continuous deployment
Contin. experimentation
Configuration mgmt.
Monitoring
Versioning
Big data
DevOps, MLOps

Teams and process: Data science vs software eng. workflows, interdisciplinary teams, collaboration points, technical debt

Responsible AI Engineering

Provenance,
versioning,
reproducibility

Safety

Security and
privacy

Fairness

Interpretability
and explainability

Transparency
and trust

Ethics, governance, regulation, compliance, organizational culture

Readings

Required reading: Shankar, Shreya, et al. “‘We Have No Idea How Models will Behave in Production until Production’: How Engineers Operationalize Machine Learning.’ Proceedings of the ACM on Human-Computer Interaction 8.CSCW1 (2024): 1-34.

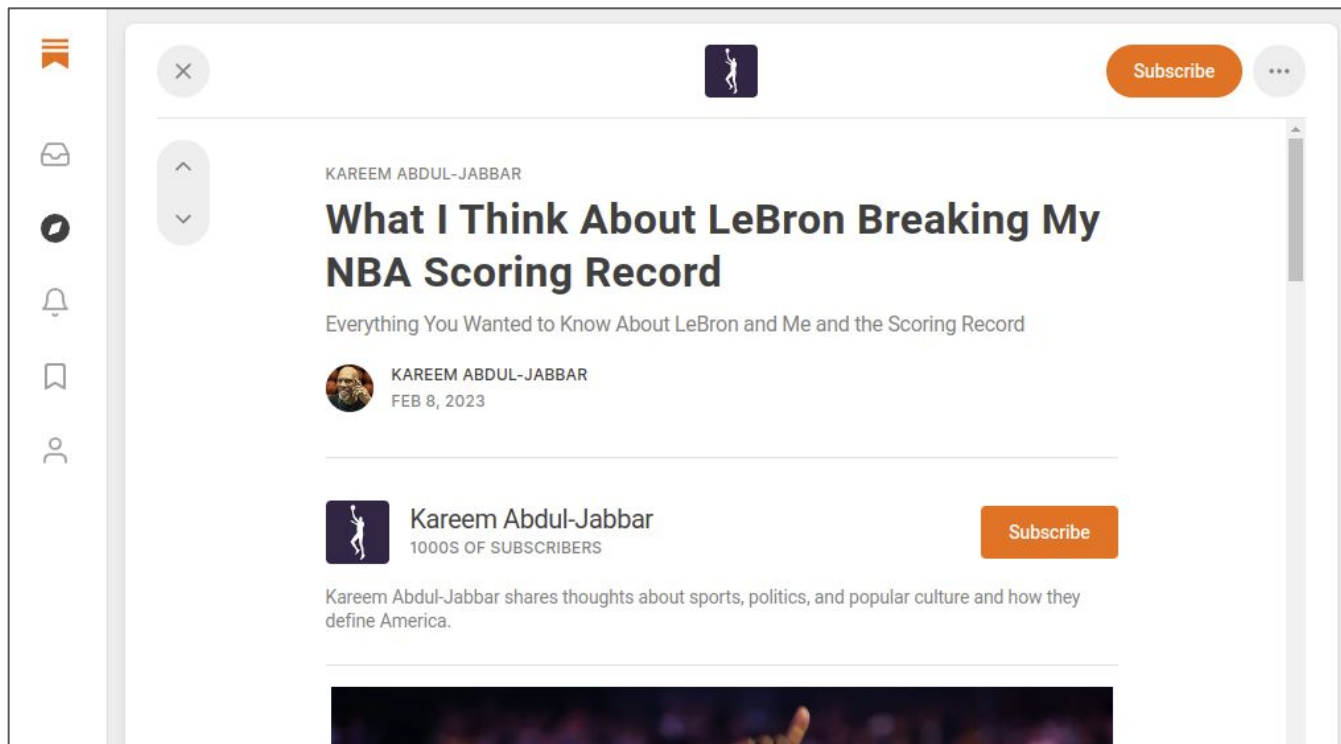
Recommended readings:

- O'Leary, Katie, and Makoto Uchida. "[Common problems with Creating Machine Learning Pipelines from Existing Code](#)." Proc. Conference on Machine Learning and Systems (MLSys) (2020).

Learning Goals

- Deploy a service for models using container infrastructure
- Automate common configuration management tasks
- Devise a monitoring strategy and suggest suitable components for implementing it
- Diagnose common operations problems
- Understand the typical concerns and concepts of MLOps

Running Example: Blogging Platform with Spam Filter



"Operations"

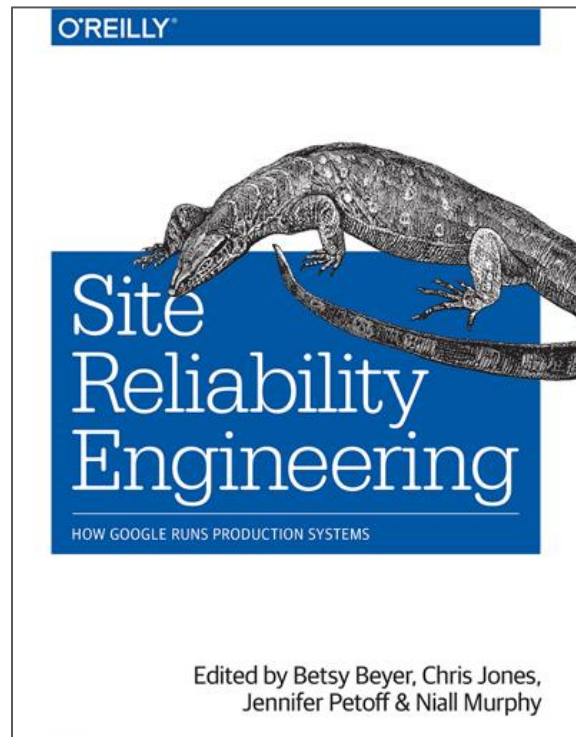
Operations

Provision and monitor the system in production, respond to problems

Avoid downtime, scale with users, manage operating costs

Heavy focus on infrastructure

Traditionally sysadmin and hardware skills



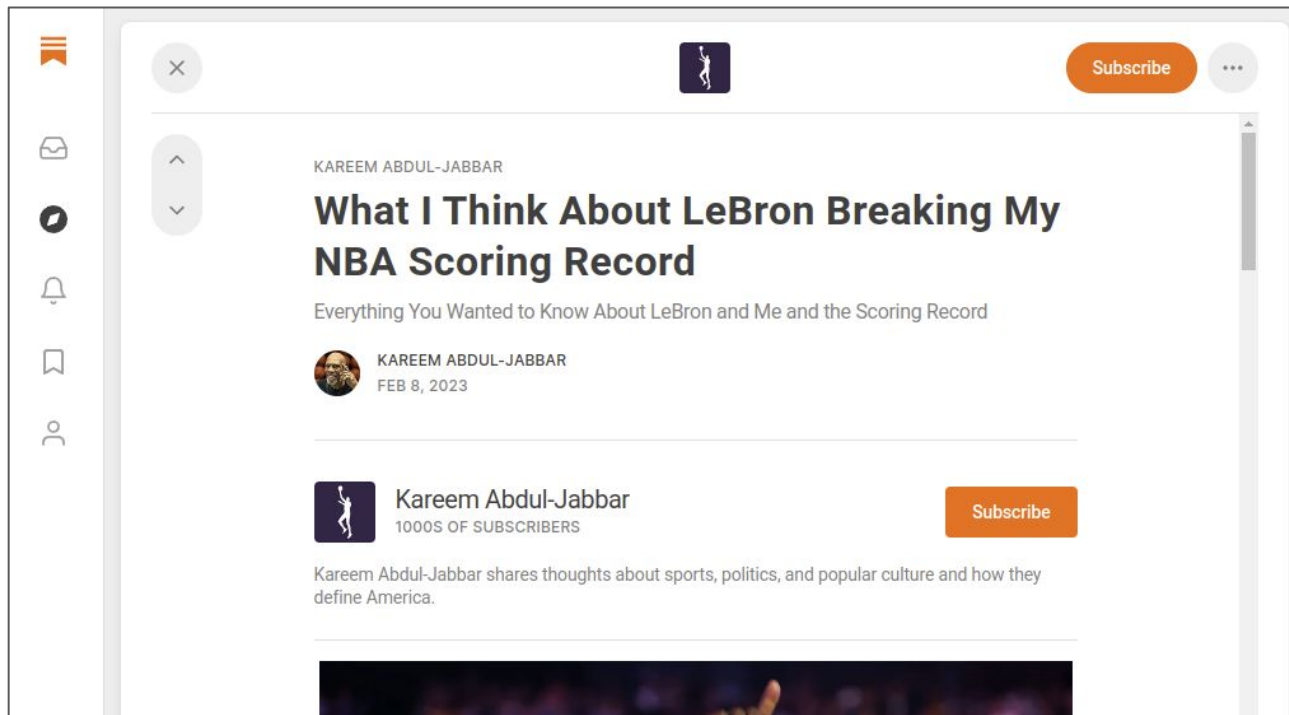
Service Level Objectives

Quality requirements in operations, each with typical measures:

- maximum latency
- minimum system throughput
- targeted availability/error rate
- time to deploy an update
- durability for storage

For the system as a whole or individual services

Example Service Level Objectives?



Operators on a Team

Operators cannot work in isolation

Rely on developers for software quality and performance

Negotiate service level agreements and budget (e.g., 99.9% vs 99.99% availability)

Risk management role (not risk avoidance)

Operations and ML

ML has distinct workloads and hardware requirements

Deep learning often pushes scale boundaries

Regular updates or learning in production

Common Themes

Observability is essential

Release management and automated deployments

Infrastructure as code and virtualization

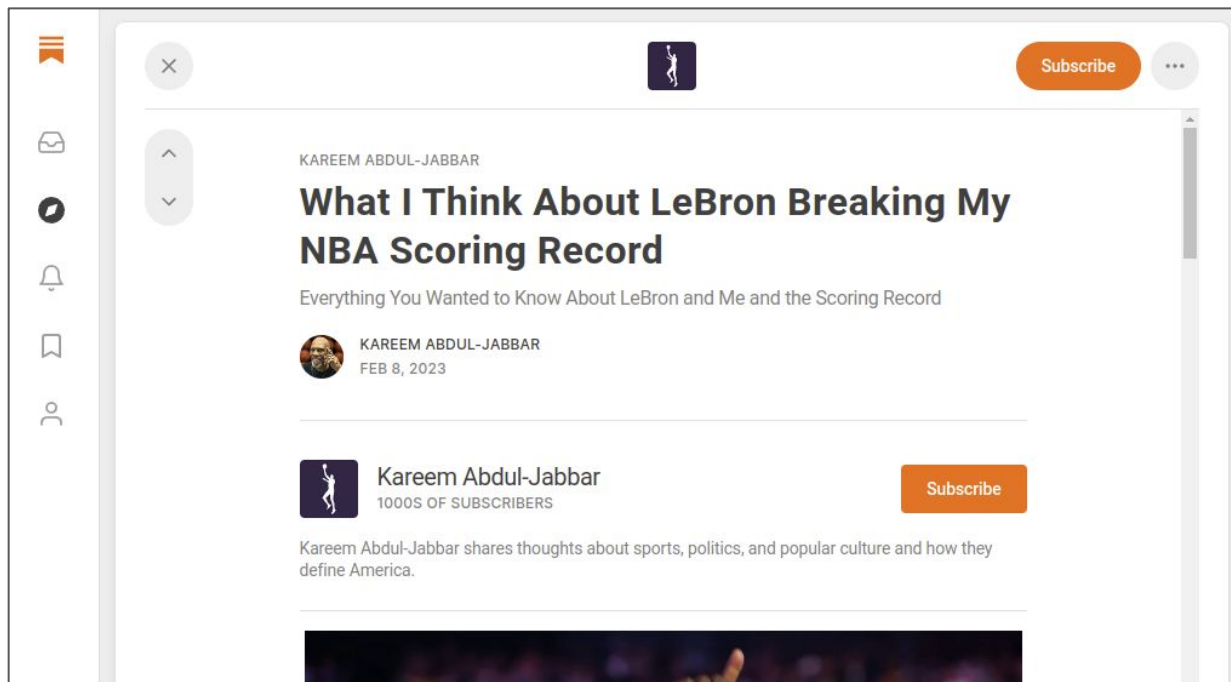
Scaling deployments

Incident response planning

Dev vs. Ops



Common Release Problems?



Common Release Problems (Examples)

- Missing dependencies
- Different compiler versions or library versions
- Different local utilities (e.g. unix grep vs mac grep)
- Database problems
- OS differences
- Too slow in real settings
- Difficult to roll back changes
- Source from many different repositories
- Obscure hardware? Cloud? Enough memory?

Developers

- Coding
- Testing, static analysis, reviews
- Continuous integration
- Bug tracking
- Running local tests and scalability experiments
- ...

Operations

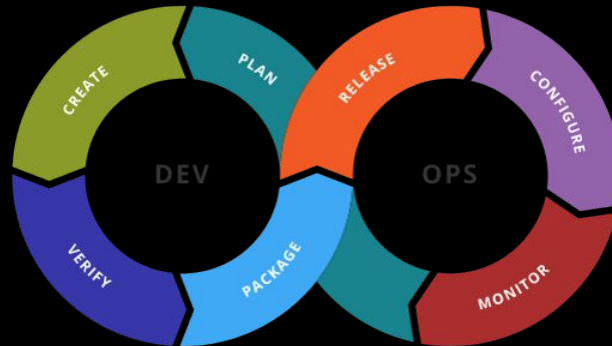
- Allocating hardware resources
- Managing OS updates
- Monitoring performance
- Monitoring crashes
- Managing load spikes, ...
- Tuning database performance
- Running distributed at scale
- Rolling back releases
- ...

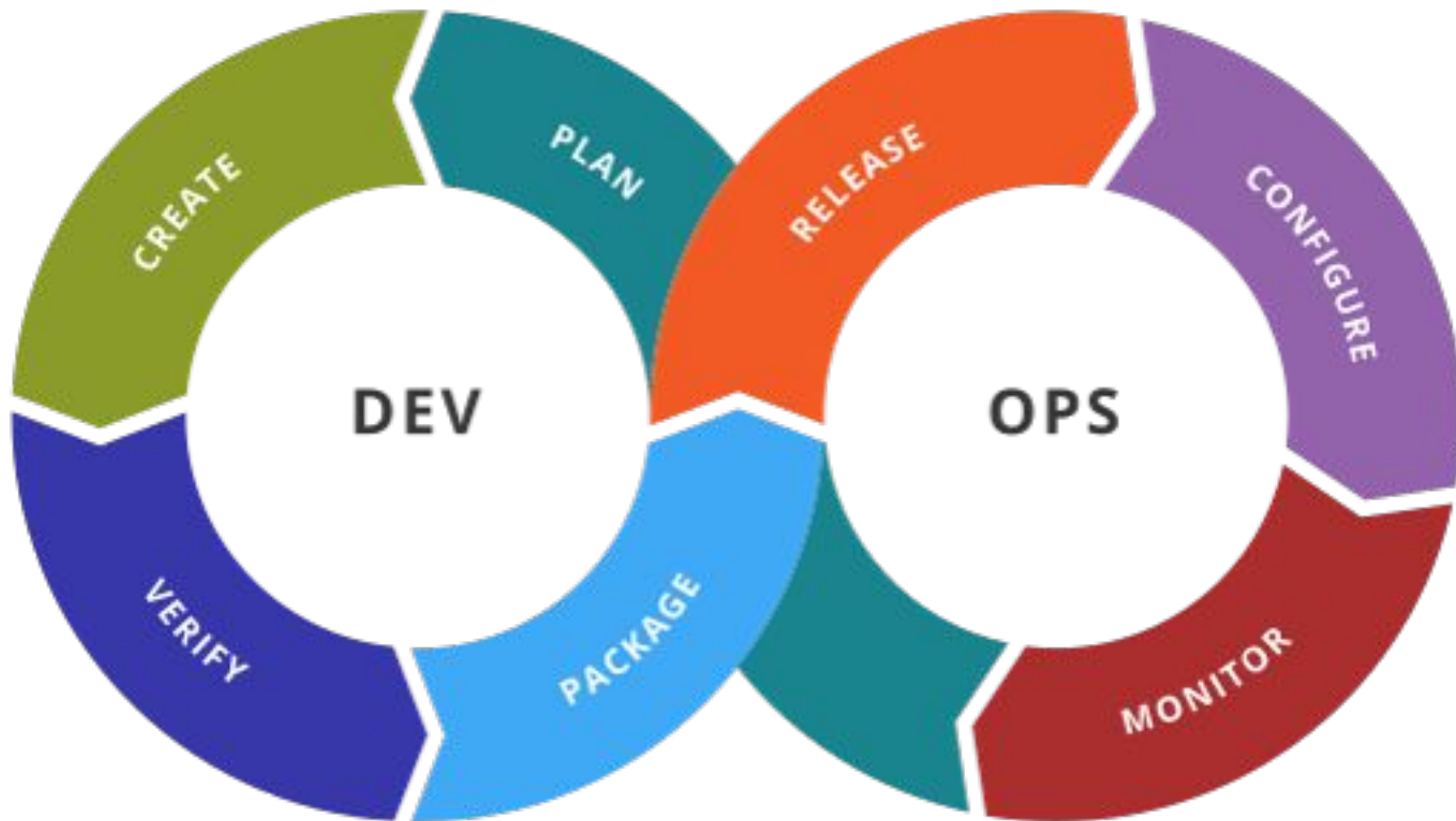
QA responsibilities in both roles

Quality Assurance does not stop in Dev

- Ensuring product builds correctly (e.g., reproducible builds)
- Ensuring scalability under real-world loads
- Supporting environment constraints from real systems (hardware, software, OS)
- Efficiency with given infrastructure
- Monitoring (server, database, Dr. Watson, etc)
- Bottlenecks, crash-prone components, ... (possibly thousands of crash reports per day/minute)

DevOps





Key ideas and principles

Better coordinate between developers and operations (collaborative)

Key goal: Reduce friction bringing changes from development into production

Considering the *entire tool chain* into production (holistic)

Documentation and versioning of all dependencies and configurations
("configuration as code")

Heavy automation, e.g., continuous delivery, monitoring

Small iterations, incremental and continuous releases

Buzz word!

Common Practices

All configurations in version control

Test and deploy in containers

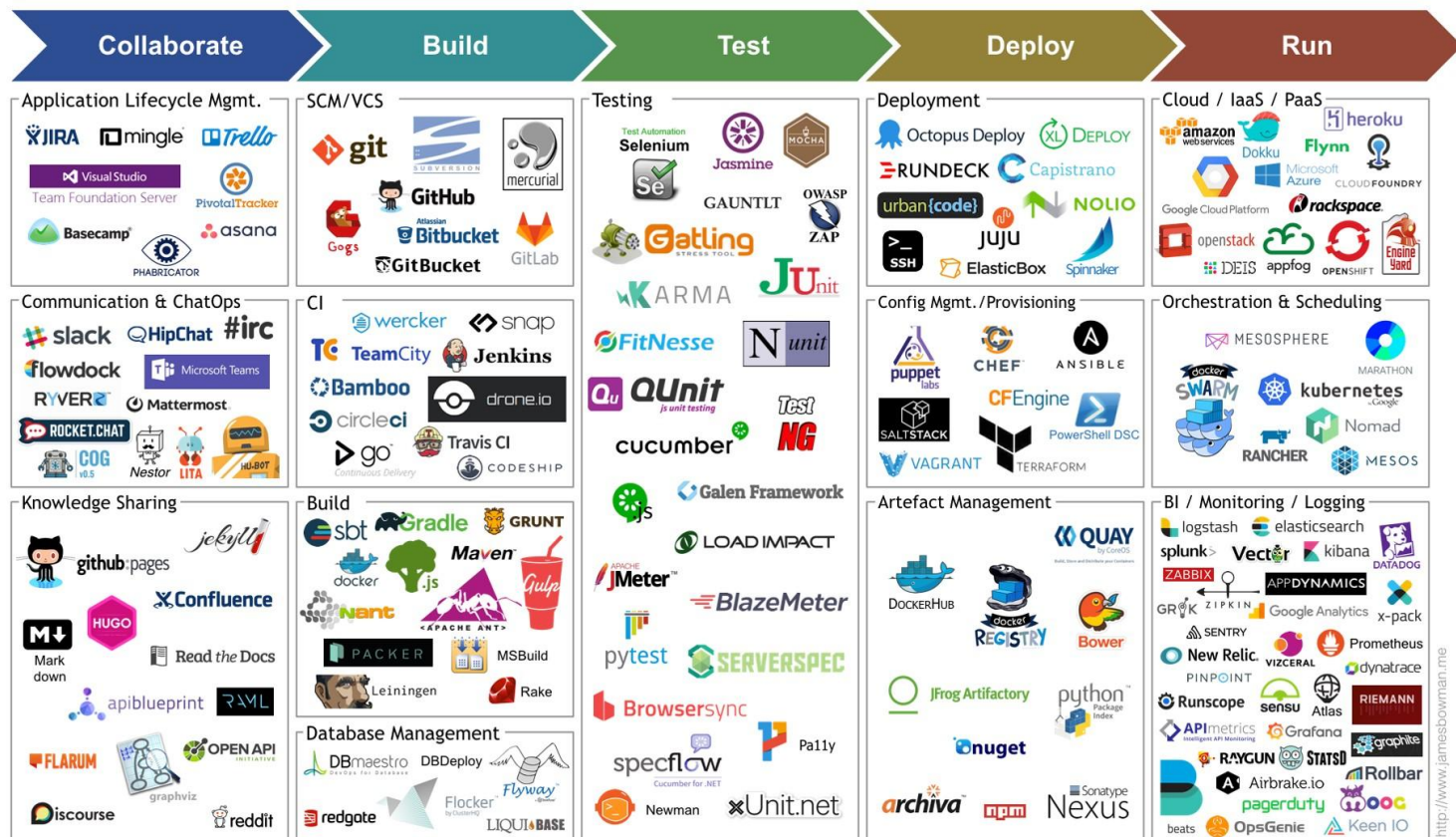
Automated testing, testing, testing, ...

Monitoring, orchestration, and automated actions in practice

Microservice architectures

Release frequently

Heavy tooling and automation



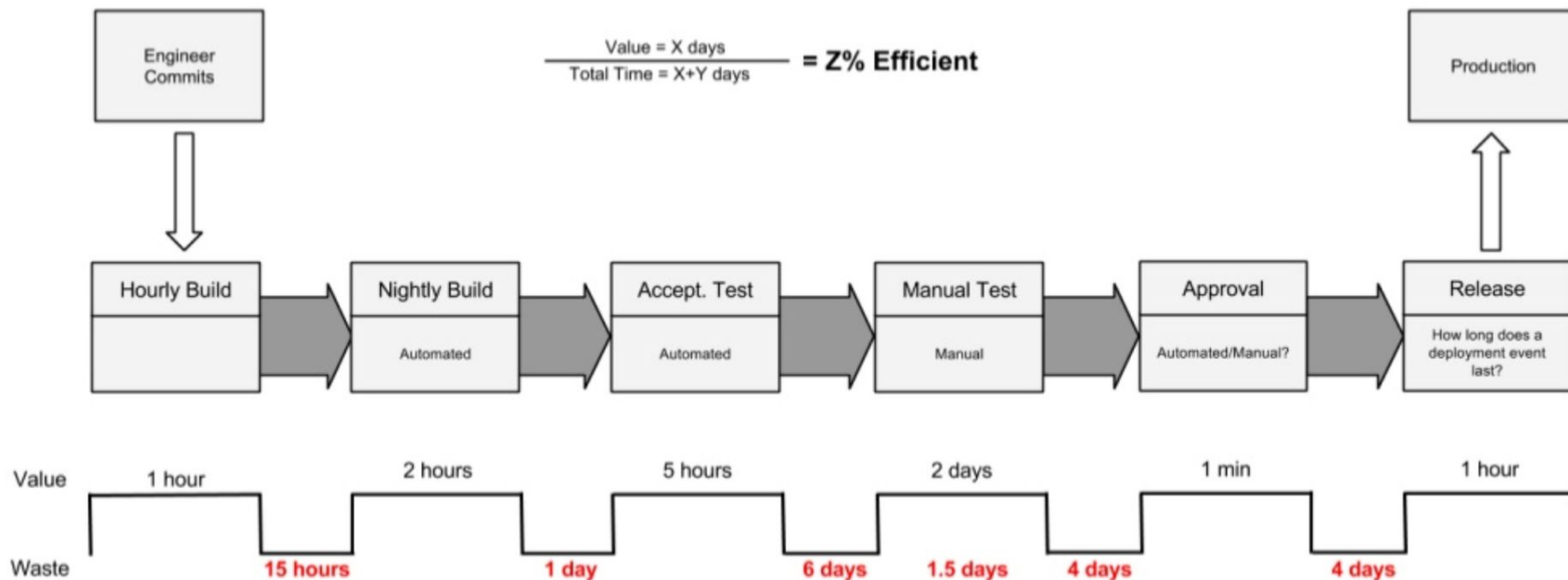
Heavy tooling and automation -- Examples

- Infrastructure as code — Ansible, Terraform, Puppet, Chef
- CI/CD — Jenkins, TeamCity, GitLab, Shippable, Bamboo, Azure DevOps
- Test automation — Selenium, Cucumber, Apache JMeter
- Containerization — Docker, Rocket, Unik
- Orchestration — Kubernetes, Swarm, Mesos
- Software deployment — Elastic Beanstalk, Octopus, Vamp
- Measurement — Datadog, DynaTrace, Kibana, NewRelic, ServiceNow

Continuous Delivery

Manual Release Pipelines

Clip slide



Continuous Integr.

- Automate tests after commit
- Independent test infrastructure

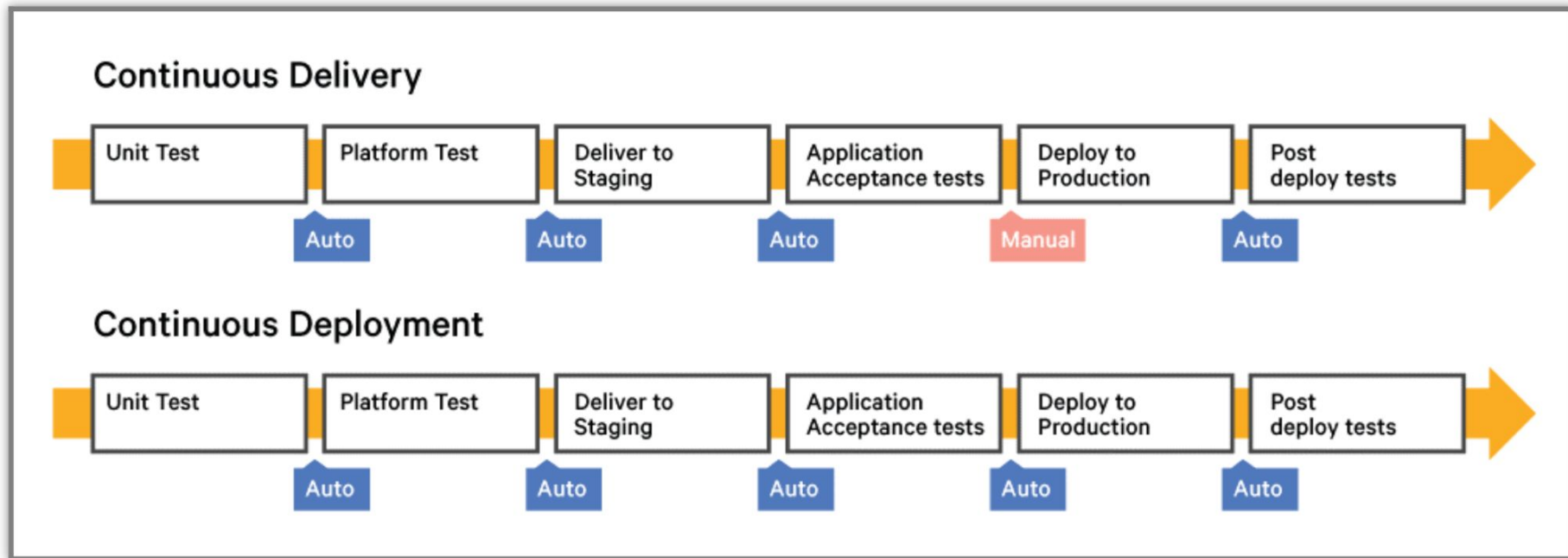
Continuous Delivery

- Full automation from commit to deployable container
- Heavy focus on testing, reproducibility and rapid feedback, creates transparency

Continuous Deployment

- Full automation from commit to deployment
- Empower developers, quick to production
- Encourage experimentation and fast incremental changes
- Commonly integrated with monitoring and canary releases

Automate Everything



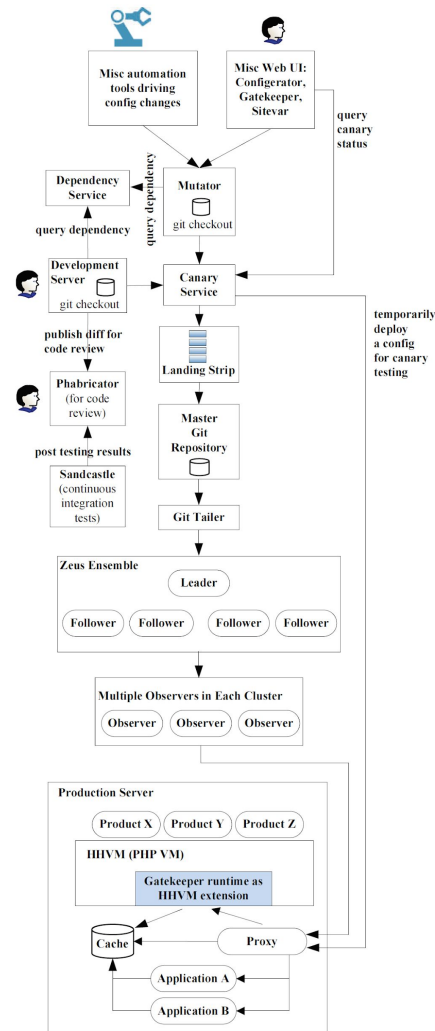
Example: Facebook Tests for Mobile Apps

- Unit tests (white box)
- Static analysis (null pointer warnings, memory leaks, ...)
- Build tests (compilation succeeds)
- Snapshot tests (screenshot comparison, pixel by pixel)
- Integration tests (black box, in simulators)
- Performance tests (resource usage)
- Capacity and conformance tests (custom)

Release Challenges for Mobile Apps

- Large downloads
- Download time at user discretion
- Different versions in production
- Pull support for old releases?
- Server side releases silent and quick, consistent
- -> App as container, most content + layout from server

Real-world pipelines are complex



Containers and Configuration Management

Containers

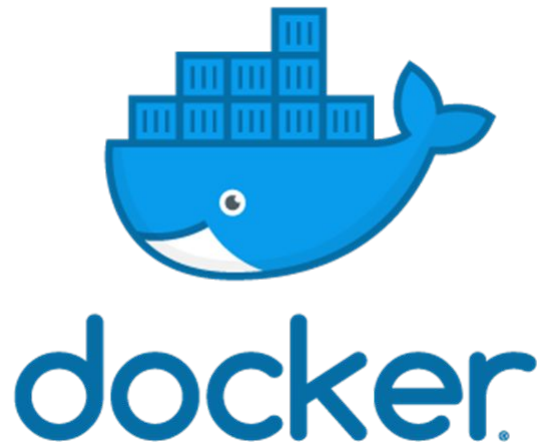
Lightweight virtual machine

Contains entire runnable software,
incl. all dependencies and
configurations

Used in development and production

Sub-second launch time

Explicit control over shared disks and
network connections



Docker Example

```
FROM ubuntu:latest
MAINTAINER ...
RUN apt-get update -y
RUN apt-get install -y python-pip python-dev build-essential
COPY . /app
WORKDIR /app
RUN pip install -r requirements.txt
ENTRYPOINT ["python"]
CMD ["app.py"]
```

Common configuration management questions

What runs where?

How are machines connected?

What (environment) parameters does software X require?

How to update dependency X everywhere?

How to scale service X?

Ansible Examples

- Software provisioning, configuration mgmt., and deployment tool
- Apply scripts to many servers

```
[webservers]
```

```
web1.company.org  
web2.company.org  
web3.company.org
```

```
[dbservers]
```

```
db1.company.org  
db2.company.org
```

```
[replication_servers]
```

```
...
```

```
# This role deploys the mongod processes and sets up the replication set.  
- name: create data directory for mongodb  
  file: path={{ mongodb_datadir_prefix }}/mongo-{{ inventory_hostname }} state=directory owner=mongod  
  group=mongod  
  delegate_to: '{{ item }}'  
  with_items: groups.replication_servers  
  
- name: create log directory for mongodb  
  file: path=/var/log/mongo state=directory owner=mongod group=mongod  
  
- name: Create the mongodb startup file  
  template: src=mongod.j2 dest=/etc/init.d/mongod-{{ inventory_hostname }} mode=0655  
  delegate_to: '{{ item }}'  
  with_items: groups.replication_servers  
  
- name: Create the mongodb configuration file  
  template: src=mongod.conf.j2 dest=/etc/mongod-{{ inventory_hostname }}.conf  
  delegate_to: '{{ item }}'  
  with_items: groups.replication_servers  
  
- name: Copy the keyfile for authentication  
  copy: src=secret dest={{ mongodb_datadir_prefix }}/secret owner=mongod group=mongod mode=0400
```

Puppet Example

Declarative specification,
can be applied to many machines

<https://www.digitalocean.com/community/tutorials/configuration-management-101-writing-puppet-manifests>

Machine Learning in Production • Christian Kaestner & Claire Le Goues, Carnegie Mellon Univer

```
$doc_root = "/var/www/example"

exec { 'apt-get update':
  command => '/usr/bin/apt-get update'
}

package { 'apache2':
  ensure  => "installed",
  require => Exec['apt-get update']
}

file { $doc_root:
  ensure => "directory",
  owner  => "www-data",
  group  => "www-data",
  mode   => 644
}

file { "$doc_root/index.html":
  ensure => "present",
  source =>
    "puppet:///modules/main/index.html",
  require => File[$doc_root]
}

file {
  "/etc/apache2/sites-available/000-default.conf":
    ensure => "present",
    content => template("main/vhost.erb"),
    notify  => Service['apache2'],
    require => Package['apache2']
}

service { 'apache2':
  ensure => running.
```

Container Orchestration with Kubernetes

Manages which container to deploy to which machine

Launches and kills containers depending on load

Manage updates and routing

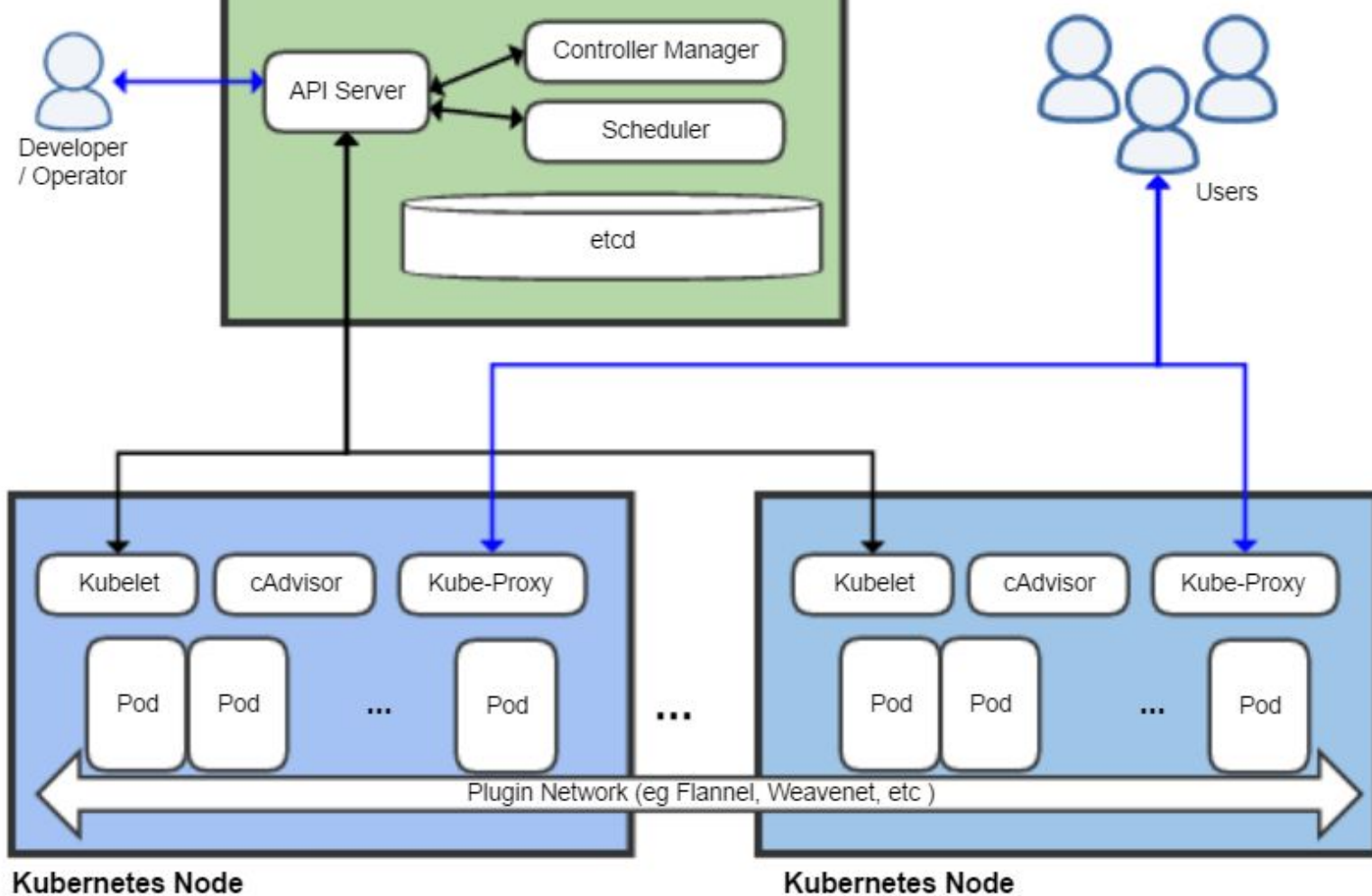
Automated restart, replacement, replication, scaling

Kubernetes master controls many nodes

Substantial complexity and learning curve

Kubernetes Master

CC BY-SA 4.0 [Khtan66](#)





Monitoring

Monitor server health

Monitor service health

Monitor telemetry

Collect and analyze measures or log files

Dashboards and triggering automated decisions

Many tools, e.g., Grafana as dashboard, Prometheus for metrics, Loki + ElasticSearch for logs

Push and pull models

The DevOps Mindset

Consider the entire process and tool chain holistically

Automation, automation, automation

Elastic infrastructure

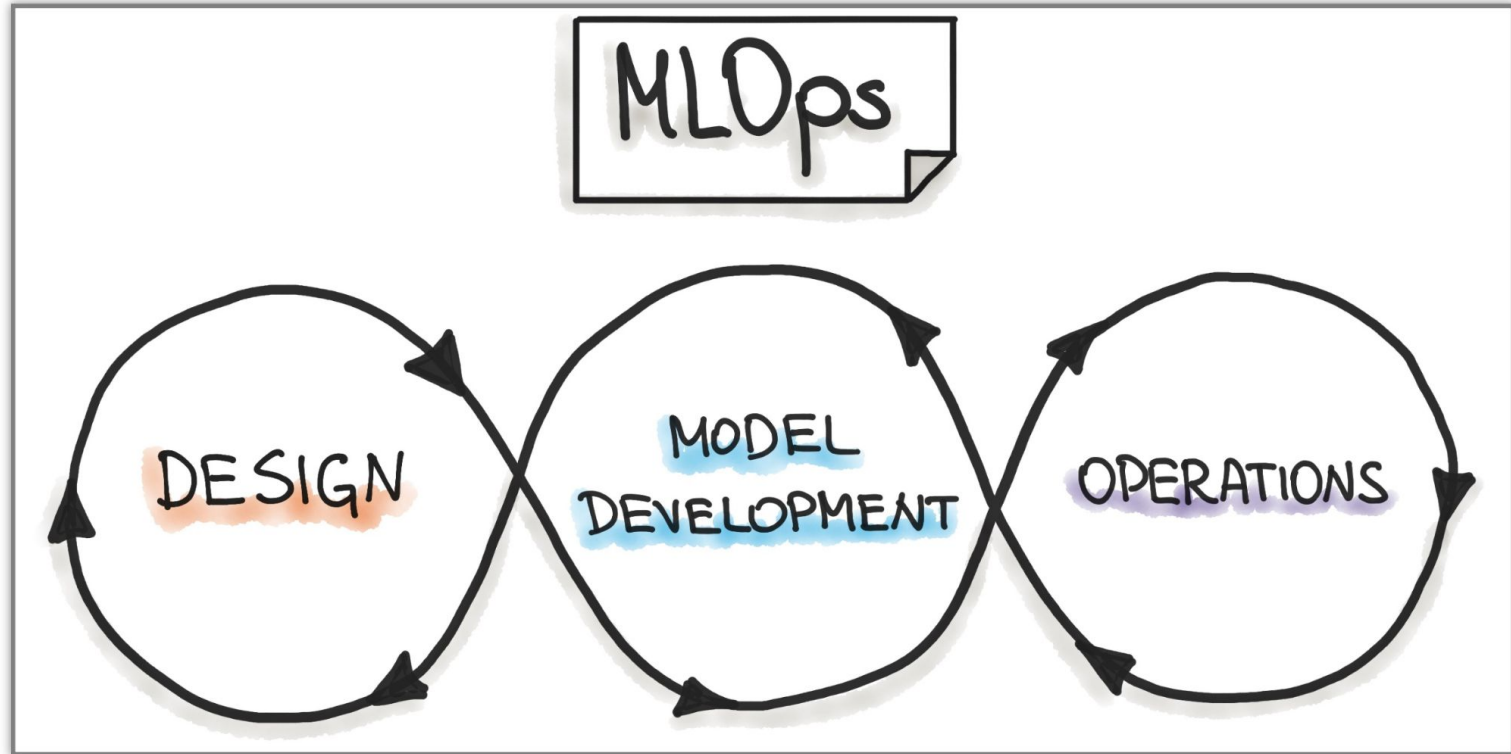
Document, test, and version everything

Iterate and release frequently

Emphasize observability

Shared goals and responsibilities

MLOps



On Terminology



- **MLOps:** Collaboration and communication between data scientists and operators, e.g.,
 - Automate model deployment
 - Model training and versioning infrastructure
 - Model deployment and monitoring
- **AIOps:** Using AI/ML to make operations decision, e.g. in a data center
- **DataOps:** Data analytics, often business setting and reporting
 - Infrastructure to collect data (ETL) and support reporting
 - Combines agile, DevOps, Lean Manufacturing ideas

MLOps Overview

Integrate ML artifacts into software release process, unify process (i.e., DevOps extension)

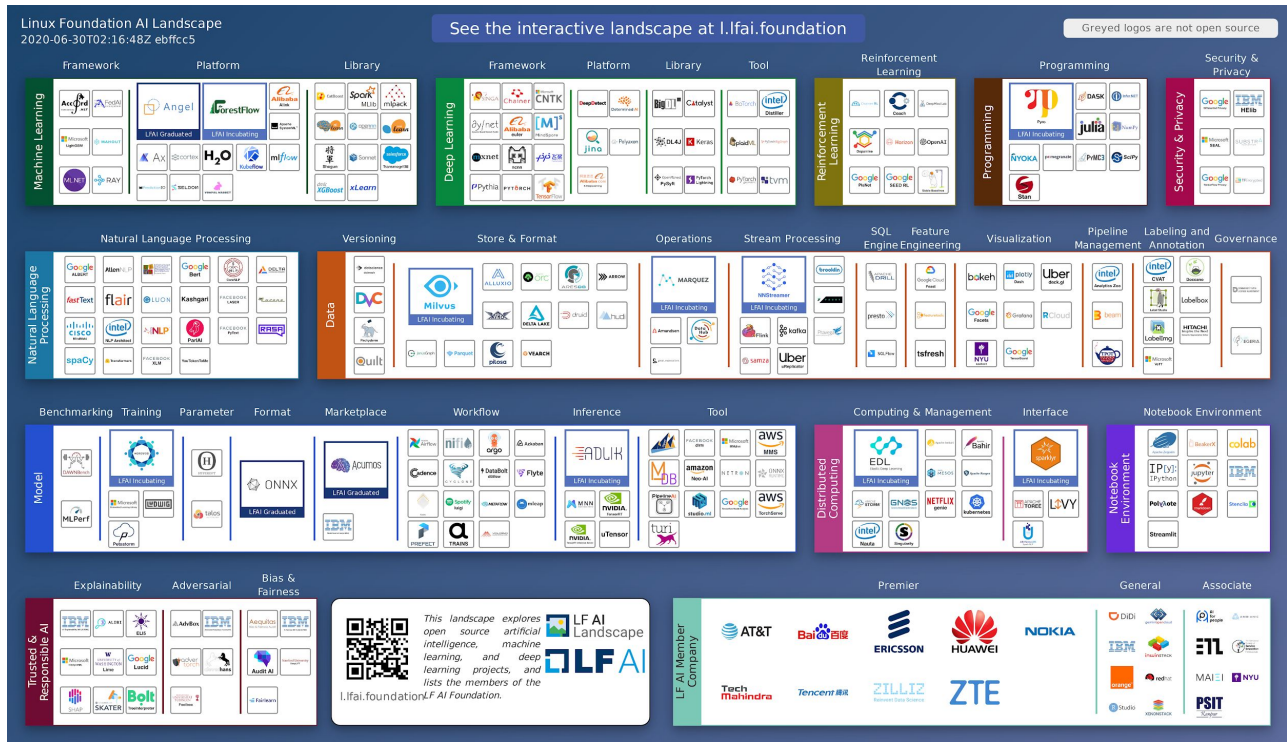
Automated data and model validation (continuous deployment)

Continuous deployment for ML models: from experimenting in notebooks to quick feedback in production

Versioning of models and datasets (more later)

Monitoring in production (discussed earlier)

Tooling Landscape LF AI



No way to put more recent versions on a slide – see

<https://landscape.lfai.foundation/?group=projects-and-products&view-mode=grid>

MLOps Goals and Principles

Like DevOps: Automation, testing, holistic, observability, teamwork

Supporting frequent experimentation, rapid prototyping, and constant iteration – with flexibility and confidence

MLOps Tools -- Examples

- Model registry, versioning and metadata: MLFlow, Neptune, ModelDB, WandB, ...
- Model monitoring: Fiddler, Hydrosphere
- Data pipeline automation and workflows: DVC, Kubeflow, Airflow
- Model packaging and deployment: BentoML, Cortex
- Distributed learning and deployment: Dask, Ray, ...
- Feature store: Feast, Tecton
- Integrated platforms: Sagemaker, Valohai, ...
- Data validation: Cerberus, Great Expectations, ...

Long lists, e.g., <https://github.com/kelvins/awesome-mlops>

MLOps Common Goals

Enable experimentation with data and models, small incremental changes; hide complexity from data scientists

Automate (nuanced) model validation (like CI) and integrate with testing in production (monitoring)

Dynamic view of constantly evolving training and test data; invest in data validation

Version data, models; track experiment results

Recall: DevOps Mindset

Consider the entire process and tool chain holistically

Automation, automation, automation

Elastic infrastructure

Document, test, and version everything

Iterate and release frequently

Emphasize observability

Shared goals and responsibilities

Breakout: MLOps Goals

For the blog spam filter scenario, consider DevOps and MLOps infrastructure (CI, CD, containers, config. mgmt, monitoring, model registry, pipeline automation, feature store, data validation, ...)

As a group, tagging group members, post to #lecture:

- *Which DevOps or MLOps goals to prioritize?*
- *Which tools to explore/invest in first?*

Incident Response Planning

Mistakes will Happen. Be Prepared

Even with careful anticipation and mitigation, mistakes will happen

Anticipated or not

ML as unreliable component raises risks

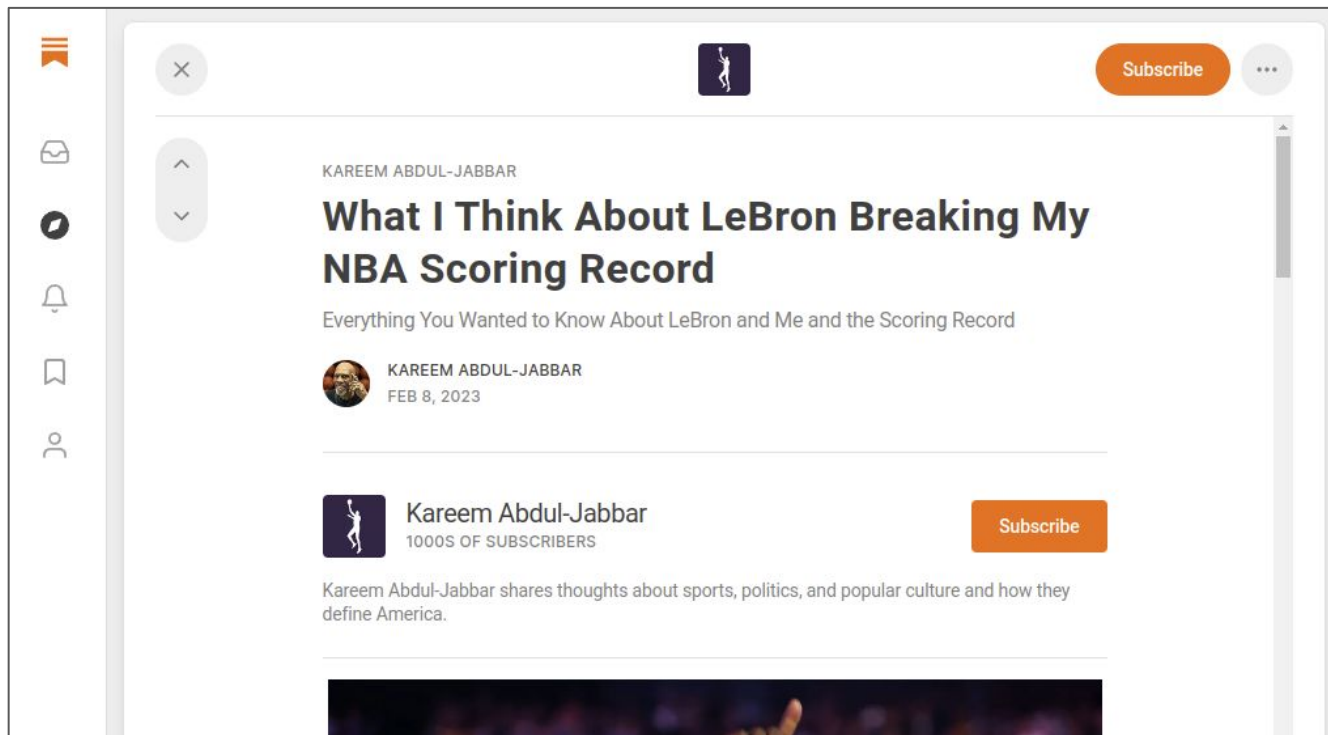
Design mitigations help avoid anticipated mistakes

Incident response plan prepares for unanticipated or unmitigated mistakes

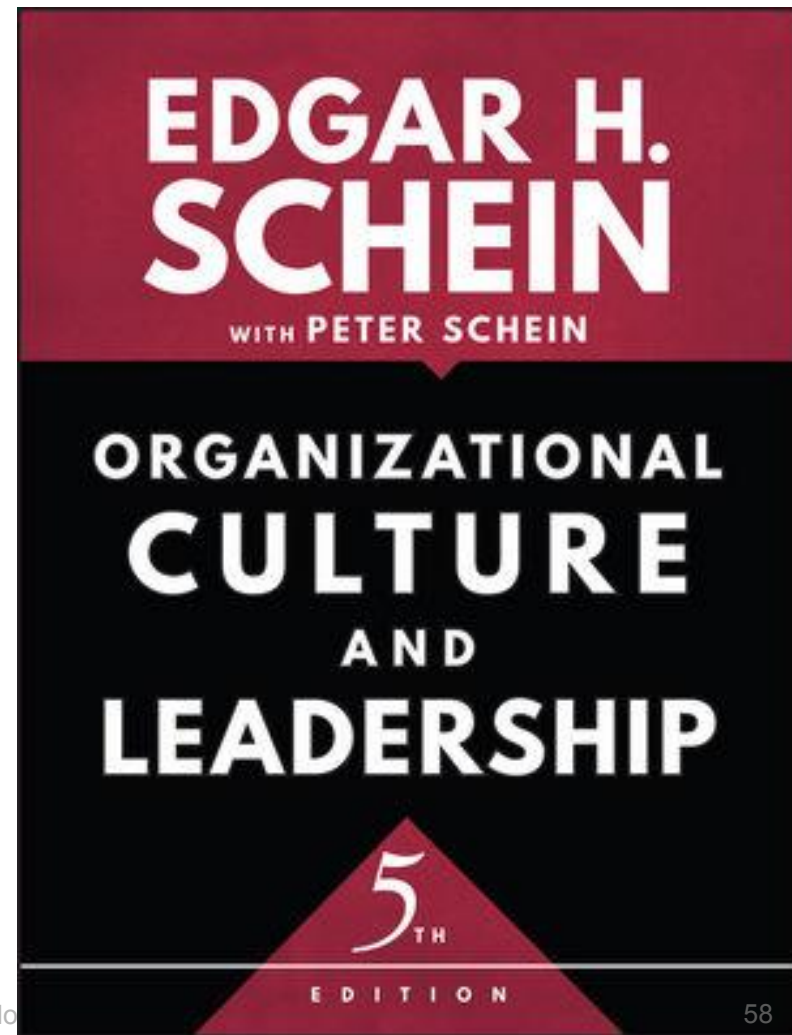
Incident Response Plan

- Provide contact channel for problem reports
- Have expert on call
- Design process for anticipated problems, e.g., rollback, reboot, takedown
- Prepare for recovery
- Proactively collect telemetry
- Investigate incidents
- Plan public communication (responsibilities)

Incident Resp. Plan for Blog's Spam Filter?



Excursion: Organizational Culture



Organizational Culture

“this is how we always did things”

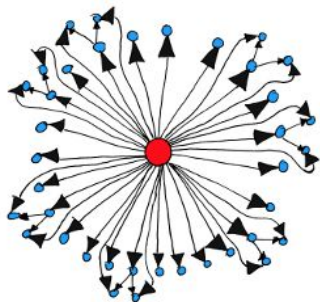
Implicit and explicit assumptions and rules guiding behavior

Often grounded in history, very difficult to change

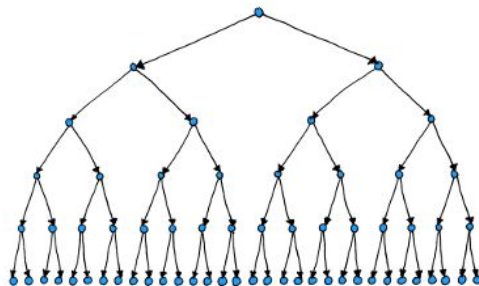
Examples:

- “Move fast and break things”
- “Privacy first”
- “Development opportunities for all employees”

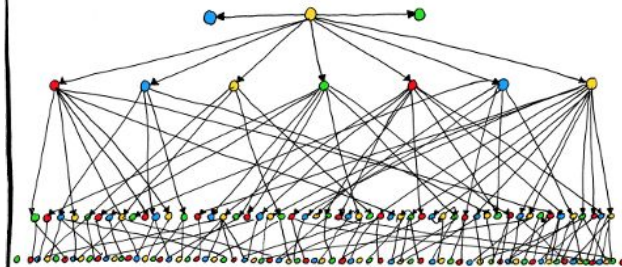
APPLE



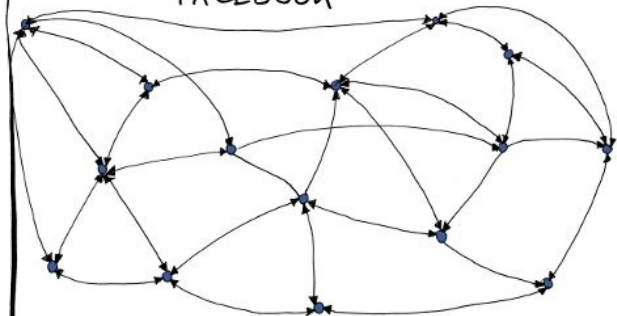
AMAZON



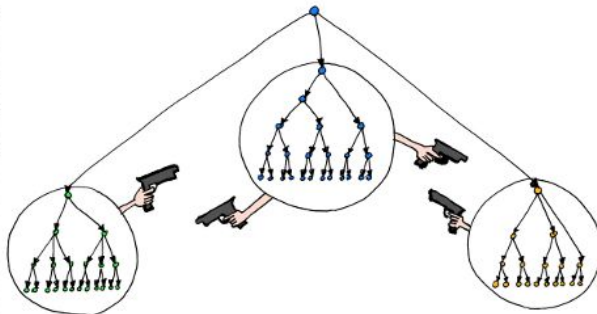
GOOGLE



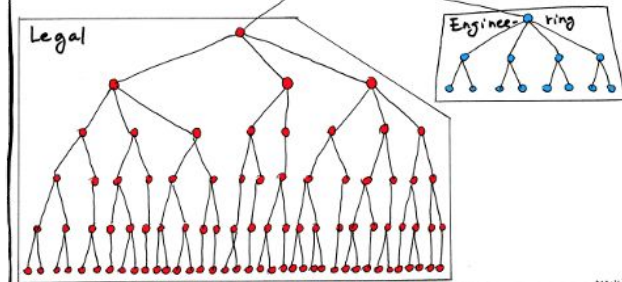
FACEBOOK



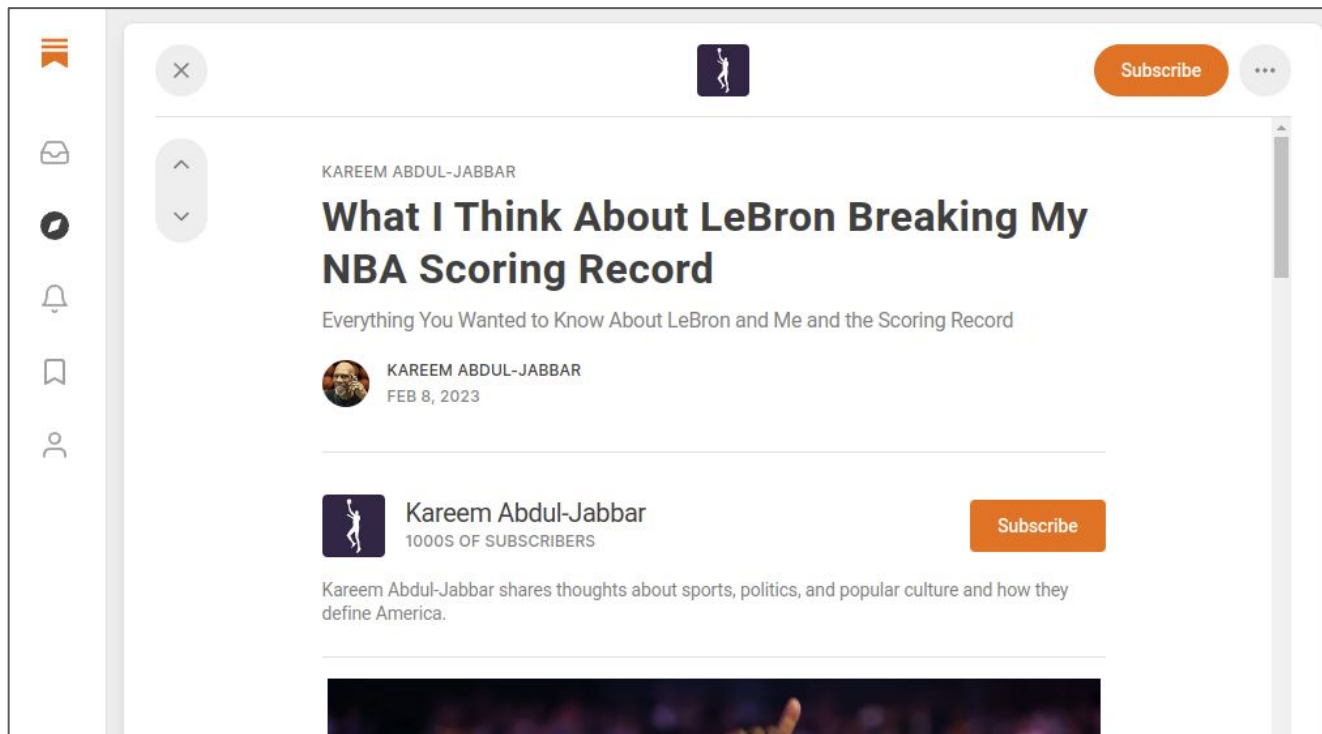
MICROSOFT



ORACLE



Organizational Culture



Levels of Organizational Culture

Artifacts -- What we see

- Behaviors, systems, processes, policies

Espoused Values -- What we say

- Ideals, goals, values, aspirations

Basic assumptions -- What we believe

- Underlying assumptions, "old ways of doing things", unconsciously taken for granted

Iceberg models: Only artifacts and espoused values visible, but practices driven by invisible basic assumptions

Culture Change

Changing organizational culture is very difficult

Top down: espoused values, management buy in, incentives

Bottom up: activism, show value, spread

Examples of success of failure stories?

MLOps Culture

Dev with Ops instead of Dev vs Ops

A culture of collaboration, joint goals, joint responsibilities

Artifacts: Joint tools, processes

Underlying assumptions: Devs provide production-ready code; Ops focus on value, automation is good, observability is important, ...

Resistance to DevOps Culture?

From "us vs them" to blameless culture -- How?

Introduction of new tools and processes -- Disruptive? Costly?

Competing with current tasks? Who wants to write tests?

Future benefits from rapid feedback and telemetry -- Unrealistic?

Automation and shifting responsibilities -- Hiring freeze and layoffs?

Past experience with poor adoption -- All costs, no benefits? Compliance only?

Successful DevOps Adoption

Need supportive management; typically driven by advocacy of individuals, convincing colleagues and management

Education to generate buy-in

Experts and consultants can help with initial costly transition

Demonstrate benefits on small project, promote afterward

Focus on key bottlenecks, over perfect adoption (e.g., prioritize experimentation, test automation, rapid feedback with telemetry)

Summary

Plan for change, plan for operations

Operations requirements: service level objectives

DevOps integrations development and operations tasks with joint goals and tools

- Heavy automation

- Continuous integration and continuous delivery

- Containers and configuration management

- Monitoring

MLOps extends this to operating pipelines and deploying models

Organizational culture is slow and difficult to change

Further Reading

- Shankar, Shreya, et al. “[‘We Have No Idea How Models will Behave in Production until Production’: How Engineers Operationalize Machine Learning](#).” Proceedings of the ACM on Human-Computer Interaction 8.CSCW1 (2024): 1-34.
- <https://ml-ops.org/>
- Beyer, Betsy, Chris Jones, Jennifer Petoff, and Niall Richard Murphy. [Site reliability engineering: How Google runs production systems](#). O’Reilly, 2016.
- Kim, Gene, Jez Humble, Patrick Debois, John Willis, and Nicole Forsgren. [The DevOps Handbook: How to Create World-Class Agility, Reliability, & Security in Technology Organizations](#). IT Revolution, 2nd ed, 2021.
- Treveil, Mark, Nicolas Omont, Clément Stenac, Kenji Lefevre, Du Phan, Joachim Zentici, Adrien Lavoillotte, Makoto Miyazaki, and Lynn Heidmann. [Introducing MLOps: How to Scale Machine Learning in the Enterprise](#). O’Reilly, 2020.
- Luz, Welder Pinheiro, Gustavo Pinto, and Rodrigo Bonifácio. “[Adopting DevOps in the real world: A theory, a model, and a case study](#).” Journal of Systems and Software 157 (2019): 110384.
- Schein, Edgar H. *Organizational culture and leadership*. 5th ed. John Wiley & Sons, 2016.